01_Ansible Lab results

The results file contains only a brief subset of the overall lab and provides limited insight into the setup process and final outcome. While it is suitable for a quick overview, readers seeking a deeper understanding of the implementation should refer to the other files included in the lab.

Ansible - DHCP setup playbook

```
GNU nano 7.2                                                    dhcp.yml
---
- name: Automated DHCP installation
  hosts: dhcp_server
  become: yes

  vars:
    interface: enp0s9
    network: 192.168.100.0
    netmask: 255.255.255.0
    range_start: 192.168.100.1
    range_end: 192.168.100.254
    gateway: 192.168.100.1
    dns: 8.8.8.8

  tasks:
    - name: Install DHCP server
      apt:
        name: isc-dhcp-server
        state: present
        update_cache: yes

    - name: Configure default network interface (where dhcp will work)
      lineinfile:
        path: /etc/default/isc-dhcp-server
        regexp: '^INTERFACESv4='
        line: "INTERFACESv4=\"{{ interface }}\""

    - name: Change DHCP configuration
      copy:
        dest: /etc/dhcp/dhcpd.conf
        content: |
          default-lease-time 600;
          max-lease-time 7200;
          authoritative;

         subnet {{ network }} netmask {{ netmask }} {
           range {{ range_start }} {{ range_end }};
           option routers {{ gateway }};
           option domain-name-servers {{ dns }};
         }

    - name: Restart DHCP service
      service:
        name: isc-dhcp-server
        state: restarted
        enabled: yes
```

Ansible – DHCP setup process

```
administrator@ubungu:~/ansible-lab/dhcp-setup$ ansible-playbook -i inventory dhcp.yml

PLAY [Automated DHCP installation] *******************************************************************

TASK [Gathering Facts] *******************************************************************************
fatal: [192.168.0.2]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: administrator@1
password).", "unreachable": true}

PLAY RECAP *******************************************************************************************
192.168.0.2                : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescued=0    ignored=0

administrator@ubungu:~/ansible-lab/dhcp-setup$ ansible-playbook -i inventory dhcp.yml --ask-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Automated DHCP installation] *******************************************************************

TASK [Gathering Facts] *******************************************************************************
ok: [192.168.0.2]

TASK [Install DHCP server] ***************************************************************************
ok: [192.168.0.2]

TASK [Configure default network interface (where dhcp will work)] ************************************
changed: [192.168.0.2]

TASK [Change DHCP configuration] *********************************************************************
changed: [192.168.0.2]

TASK [Restart DHCP service] **************************************************************************
changed: [192.168.0.2]

PLAY RECAP *******************************************************************************************
192.168.0.2                : ok=5    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

administrator@ubungu:~/ansible-lab/dhcp-setup$ _
```

Result of setup , client machine has received IP address(enp0s9) from previously setted up DHCP server by Ansible server

```
administrator@ubungu:~/ansible-lab/dhcp-setup$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:ac:ab brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 85769sec preferred_lft 85769sec
    inet6 fd17:625c:f037:2:a00:27ff:fe55:acab/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86262sec preferred_lft 14262sec
    inet6 fe80::a00:27ff:fe55:acab/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d5:b6:4b brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.1/24 brd 192.168.0.255 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed5:b64b/64 scope link
       valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ba:ce:28 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.2/24 metric 100 brd 192.168.100.255 scope global dynamic enp0s9
       valid_lft 419sec preferred_lft 419sec
    inet6 fe80::a00:27ff:feba:ce28/64 scope link
       valid_lft forever preferred_lft forever
administrator@ubungu:~/ansible-lab/dhcp-setup$
```